Finite-volume scalar spectral constructive Gibbs measure

TECT verification-first repository · claim A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE

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1. Purpose and scope

This note constructs the non-perturbative finite-volume Gibbs measure for the real scalar Brazovskii slice. It is deliberately a separate claim from the perturbative A3 correlator theorem and from the P3 PDE-discretization theorem. The torus is

$$\mathbb{T}^3 = [0, L)^3, \qquad L = 16, \qquad |\mathbb{T}^3| = L^3. \tag{1.1}$$

Let $\alpha = 2\pi/L$, let P_N retain the Fourier modes $n \in \mathbb{Z}^3$ with $|n|_\infty \le N$, and define

$$K_n = m_{\text{sh}}^2 + Y\big(\alpha^2|n|^2 - q_0^2\big)^2, \qquad C_n = K_n^{-1}. \tag{1.2}$$

The hypotheses are

$$m_{\text{sh}}^2 > 0, \qquad Y > 0, \qquad q_0 \ge 0, \qquad \lambda \in \mathbb{R}, \qquad \gamma > 0. \tag{1.3}$$

For a real field ϕ, put

$$V(\phi) = \int_{\mathbb{T}^3} \left(\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6\right) dx. \tag{1.4}$$

Pinned-scope theorem. Let μ_0 be the centred real Gaussian measure with covariance $C = K^{-1}$. The covariance is trace class on real $L^2(\mathbb{T}^3)$. The Galerkin laws

$$d\nu_N(\phi_N) = Z_N^{-1} e^{-V(\phi_N)} d\mu_{0,N}(\phi_N), \qquad \mu_{0,N} = (P_N)_\# \mu_0, \tag{1.5}$$

are normalizable. There is a probability measure

$$d\nu(\phi) = Z^{-1} e^{-V(\phi)} d\mu_0(\phi) \tag{1.6}$$

such that the full sequence ν_N converges weakly to ν on real L^2. The lifted densities on the common Gaussian space converge in $L^1(\mu_0)$, equivalently in total variation, and every finite-degree smeared cylinder polynomial has a convergent expectation. This is cutoff removal for the declared spectral regulator; it is not an infinite-volume limit.

1 Trace class and Gaussian regularity

On the max-norm shell $|n|_\infty = m$ there are exactly

$$(2m+1)^3 - (2m-1)^3 = 24m^2 + 2 \tag{2.1}$$

modes. If $m \ge m_0 := \lceil \sqrt{2} q_0/\alpha \rceil$, then $\alpha^2|n|^2 \ge 2q_0^2$ and hence

$$K_n \ge \frac{Y}{4}\alpha^4 m^4. \tag{2.2}$$

Consequently,

$$\sum_{|n|_\infty \geq M} C_n \leq \frac{4}{Y\alpha^4} \sum_{m=M}^{\infty} \left(24m^{-2} + 2m^{-4}\right) < \infty. \tag{2.3}$$

The finitely many inner modes are harmless because $K_n \geq m_{\mathrm{sh}}^2$. Thus C is trace class and μ_0 is a Radon Gaussian probability measure on real L^2.

With the H^s weight, the shell summand is bounded by a constant times m^{2s-2}. It is summable for every $s < 1/2$; negative s already follows from L^2 support. Conversely, $K_n \leq c(1 + |n|^4)$ at large $|n|$, so the weighted covariance trace has the harmonic lower power at $s = 1/2$. Hence the covariance is trace class on H^s for $s < 1/2$, but not on $H^{1/2}$. This is the sharp Gaussian Hilbert-space threshold used here.

The low Sobolev exponent does not obstruct the local polynomial. For the common Gaussian Fourier series, the tail $X_N = (I - P_N)\phi$ has an x-independent variance

$$t_N = |\mathbb{T}^3|^{-1} \sum_{|n|_\infty > N} C_n \longrightarrow 0. \tag{2.4}$$

Since a centred real Gaussian X satisfies $\mathbb{E}X^6 = 15(\mathbb{E}X^2)^3$,

$$\mathbb{E}\int_{\mathbb{T}^3} |X_N(x)|^6 dx = 15|\mathbb{T}^3| t_N^3 \longrightarrow 0. \tag{2.5}$$

Therefore $P_N\phi \to \phi$ in $L^6(\Omega \times \mathbb{T}^3)$. In particular, $\phi \in L^6$ almost surely and (1.4) is a finite random variable. This direct Gaussian-moment argument, not an invalid $H^{1/2-} \to L^6$ Sobolev embedding, defines the interaction.

2 Stability and partition functions

Write $a = \max(-\lambda, 0)$. For $x = \phi^2 \geq 0$, the possibly negative part of the potential is minimized at $x = a/\gamma$. Direct substitution gives the sharp pointwise bound

$$\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6 \geq -\frac{a^3}{12\gamma^2} = -c_*. \tag{3.1}$$

It follows immediately that

$$e^{-V(P_N\phi)} \leq e^{c_*|\mathbb{T}^3|}, \qquad Z_N \leq e^{c_*|\mathbb{T}^3|}. \tag{3.2}$$

Let

$$\sigma_*^2 = |\mathbb{T}^3|^{-1} \operatorname{Tr} C. \tag{3.3}$$

Translation invariance and the real Gaussian moments $\mathbb{E}X^4 = 3\sigma^4$, $\mathbb{E}X^6 = 15\sigma^6$ give

$$\mathbb{E}V(P_N\phi) \leq |\mathbb{T}^3| \left(\frac{3\lambda_+}{4}\sigma_*^4 + \frac{5\gamma}{2}\sigma_*^6\right) = B_* < \infty. \tag{3.4}$$

Jensen's inequality has the required direction:

$$Z_N = \mathbb{E}e^{-V(P_N\phi)} \geq e^{-\mathbb{E}V(P_N\phi)} \geq e^{-B_*} > 0. \tag{3.5}$$

Equations (3.2) and (3.5) uniformly enclose every Galerkin partition function.

3 Full-sequence regulator removal

The elementary polynomial difference inequality, Holder's inequality on $\Omega \times \mathbb{T}^3$, and (2.5) yield

$$\mathbb{E}|V(P_N\phi) - V(\phi)| \longrightarrow 0. \tag{4.1}$$

For example, the sextic term is bounded by a constant times

$$(\|P_N\phi\|_{L^6}^5 + \|\phi\|_{L^6}^5)\|P_N\phi - \phi\|_{L^6}, \tag{4.2}$$

and the quartic term is easier on the finite product space. Thus $V(P_N\phi) \to V(\phi)$ in probability. By (3.1), all weights are bounded by the same deterministic constant. Bounded convergence in probability gives

$$w_N := e^{-V(P_N\phi)} \longrightarrow w := e^{-V(\phi)} \quad \text{in } L^1(\mu_0). \tag{4.3}$$

Hence $Z_N \to Z = \mathbb{E}w$. The limit is strictly positive because $w > 0$ almost surely; equivalently, (3.5) passes to the limit.

Define the lifted probability measure

$$d\widetilde{\nu}_N = Z_N^{-1} w_N d\mu_0. \tag{4.4}$$

Normalization and (4.3) imply

$$\left\| \frac{w_N}{Z_N} - \frac{w}{Z} \right\|_{L^1(\mu_0)} \longrightarrow 0, \tag{4.5}$$

which is total-variation convergence of the lifted laws. The actual finite-dimensional law is $(P_N)_\# \widetilde{\nu}_N = \nu_N$. For every bounded continuous F on L^2, $P_N\phi \to \phi$ almost surely and the normalized densities are uniformly bounded by (3.2)–(3.5). Therefore

$$\int F(\phi_N) d\nu_N = \int F(P_N\phi) d\widetilde{\nu}_N \longrightarrow \int F(\phi) d\nu. \tag{4.6}$$

This identifies the full sequence, so no unnamed subsequence or uniqueness assumption is hidden in the cutoff removal. Weak convergence also supplies tightness on the Polish space L^2.

The total-variation statement in (4.5) is only for the common-Gaussian lifted laws. The embedded finite-dimensional laws have different supports and are not claimed to converge in total variation.

4 Declared correlations

Let $\ell_1, \dots, \ell_r$ be continuous real linear functionals on L^2 and let Q be a finite-degree polynomial. The observable

$$\mathcal{O}(\phi) = Q(\ell_1(\phi), \dots, \ell_r(\phi)) \tag{5.1}$$

has Gaussian moments of every order. The uniform density bound from (3.2)–(3.5) transfers uniform integrability to $\widetilde{\nu}_N$. Since $\ell_j(P_N\phi) \to \ell_j(\phi)$ in every finite Gaussian moment,

$$\int Q(\ell_1(\phi_N), \dots, \ell_r(\phi_N)) d\nu_N \longrightarrow \int \mathcal{O}(\phi) d\nu. \tag{5.2}$$

Thus smeared cylinder correlations have regulator removal. Unsmeared coincident-point products, composite-operator renormalization, and infinite- volume correlation decay are not statements of this theorem.

5 Reproducible quantitative audit

The primary audit derives all bounds from two hash-pinned anchors. The non-importing audit reconstructs the Fourier trace with scalar loops, the shell count, the Decimal potential minimum, the Gaussian moments, and the partition bounds. The integrated verifier reruns both in a temporary directory and checks all three source hashes.

For the perturbative scalar anchor $m_{\mathrm{sh}}^2 = 0.005$, the conservative covariance-trace upper bound is 3029.399163006103 and the pointwise variance upper bound is 0.7395994050307869. For the production-local sanity anchor,

$$m_{\mathrm{sh}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475, \tag{6.1}$$

the corresponding bounds are 596.857766699839 and 0.14571722819820287. The exact stability constant from the pinned local coefficients is

$$c_* = 0.0025246087994716246. \tag{6.2}$$

These are derived sanity checks from stored inputs, not physical predictions and not restrictions of the general theorem.

The internal result is

```
PASS: primary (17/17)
PASS: independent (14/14)
ASSERTS: 31/31
A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS
```

from

```
python codes/foundations/a4_scalar_constructive_measure_verify.py
```

Any failed assertion or source-hash drift reopens the closure gate.

6 Devil's-advocate review

1. **Trace class alone defines the local sextic interaction. UPHELD as false.** Equation (2.5) separately proves Gaussian L^6 convergence and almost-sure integrability.

2. **$H^{1/2-}$ embeds into L^6 in three dimensions. UPHELD as false.** No such embedding is used; the pointwise Gaussian sixth-moment identity is the load-bearing route.

3. **A negative quartic makes the Gibbs weight unbounded. DISMISSED under the stated hypotheses.** The positive sextic gives the exact global lower bound (3.1), hence the uniform weight bound (3.2).

4. **Jensen's inequality could have been used in the wrong direction. DISMISSED.** The convex function is e^{-x}, giving $\mathbb{E}e^{-V} \geq e^{-\mathbb{E}V}$ as displayed in (3.5).

5. **Tightness only gives an unidentified subsequence. UPHELD as insufficient by itself.** The L^1 density limit (4.5) identifies the full sequence before tightness is cited.

6. **The finite-dimensional measures converge in total variation after zero-extension. UPHELD as false.** Only the lifted laws converge in total variation; the actual Galerkin laws converge weakly on L^2.

7. **Every local or composite correlation is covered. UPHELD as false.** Section 5 is restricted to finite-degree smeared cylinder polynomials.

8. **The same proof covers derivative Class-II terms. UPHELD as false.** Typical fields have only the covariance regularity of Section 2; derivative-current squares require a separate definition and possibly renormalization.

9. **Finite volume establishes a phase transition. UPHELD as false.** No thermodynamic limit, symmetry breaking, or BCC selection is included.

10. **Large finite constants are practical error bars. UPHELD as false.** The bounds certify existence and convergence, not useful cutoff sizes or Monte Carlo efficiency.

7 Falsifier and no-overclaim boundary

The claim is falsified by a non-trace-class covariance in the declared three-dimensional positive-mass q^4 scope; failure of (2.5), (3.1), or the uniform partition bounds; a failure of L^1 weight convergence; two Galerkin subsequences with different weak limits; a declared smeared cylinder correlation that does not converge; or failure of the 31-assertion verifier.

This result does not construct the full three-complex-component derivative Class-II Gibbs measure. It also excludes $\eta_{\text{shell}} \neq 0$, zero or negative shell mass, infinite volume, phase transition, a genuine finite-difference Route B limit, BCC existence or selection, and the future P5 Sector-A synthesis. Independent operator execution is required before the separate T6 conditional-theorem review.

Result footer

```
Result ID:              A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
Precise statement:      The full sharp spectral Galerkin Gibbs sequence
                        converges weakly on L2; common-Gaussian lifted
                        densities converge in L1/total variation.
Scope:                  T5 CLOSED@FINITE-VOLUME-REAL-SCALAR-SPECTRAL;
                        T3 with periods 16, m_sh^2>0, Y>0, q0>=0,
                        lambda real, gamma>0, beta=1.
Dependencies:           A1-SCALAR-ANALYTIC-BRANCH;
                        hypotheses A1-SHELL-POSITIVITY and
                        A2-H2-SEXTIC-COERCIVITY.
Evidence grade:         ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:   python codes/foundations/
                        a4_scalar_constructive_measure_verify.py
Expected output:        primary 17/17; independent 14/14; ASSERTS 31/31;
                        A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS.
Falsification gate:     Failed trace/L6/stability/partition/full-sequence/
                        correlation argument, audit, or source hash.
Tier before / after:    T3 / T5, justified as a one-shot textbook argument
                        plus two independent executable reconstructions.
No-overclaim statement: Not full derivative Class-II, infinite volume,
                        phase transition, Route B, BCC, T6/T7, or P5.
Next required action:   Independent operator execution, then scoped T6
                        review; full Class-II remains a separate project.
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